Shanmukh Upadhyayula

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Education

University of Illinois Urbana-Champaign — Urbana, IL

Bachelor of Science in Computer Science & Geographic Information Science — *May 2027 (Expected)*

- GPA: 3.95/4.0 | Involvement: ACM Project Lead in Special Interest Groups for HCI and AI
- Relevant Coursework: Applied Linear Algebra, Virtual Reality, Computer Systems, Data Structures, Algorithms & Models of Computation, Human-LLM Interaction, Spatial Analysis
- Upcoming: Computational Photography, Computer Vision, Applied Machine Learning

Experience

UIUC CyberGIS Lab — Urbana, IL

AI Intern — *September 2026 – Present*

- Benchmarking the lab's geospatial AI agent on all 50 GeoAnalystBench tasks, where it matches published LLMs on code but plans more steps than the expert on 43 of 48 tasks
- Running user tests on the live agent across dataset search, map projections, raster data, file uploads, and adversarial prompts, and reporting 4 bugs that break core features
- Improving the agent's architecture, including how it picks tools, keeps its search index current, and runs code

Semantic Segmentation Failure Analysis | **Paper** — Urbana, IL

Undergraduate Researcher — *May 2026 – August 2026*

- Trained a DeepLabV3-ResNet50 model on aerial imagery and built a tiled scanner that maps where it is likely wrong across full-size images, using only the model's own confidence and no labeled data
- Validated the error maps on 4 benchmark datasets, found the model is about half as accurate along object edges, and presented the work in an IEEE-format paper and a CyberGIS talk

Digital Diary (ACM SIG-CHI) | **Website** — Urbana, IL

Project Lead — *September 2025 – April 2026*

- Built a real-time camera pipeline with OpenCV and MediaPipe that tracks face and hand landmarks from webcam video to detect mood and gesture controls, running as its own Docker service
- Led a 15-person team building a Next.js journaling app with a guided mode that pulls in Google Calendar, weather data, and Claude AI

UIC Electronic Visualization Laboratory | **Repo** — Chicago, IL

Software Engineering Intern — *June 2025 – August 2025*

- Built a Pandas merge tool that cut processing time by about 30% across hundreds of Chicago urban data pipelines
- Added dataset merging and multi-tab data views to Curio, a visual data-pipeline tool used by researchers

Projects

Urban AR GIS: Hand-Tracked 3D City Viewer | *Python* | Repo — March 2026

- Built a webcam AR viewer where MediaPipe hand tracking lets users zoom, pan, and grab 973 3D campus buildings with gestures, with tracking on its own thread at about 30 FPS
- Wrote ModernGL/GLSL shaders for a 60 FPS renderer with Blinn-Phong lighting, gamma correction on high-resolution satellite imagery, and atmospheric haze

Off the Clock: Standalone VR Game | *C#, Unity XR* | *Virtual Reality Course Project* — April 2026 – May 2026

- Built motion controls and a combat system for sword fighting, laser guns, and mech flight, using controller tracking data and sphere-cast sweeps so fast sword swings register hits reliably
- Tested Android builds on Meta Quest 3 against a 90 FPS target, adding an in-headset overlay for frame time, draw calls, and memory use to catch performance drops on the device
- Debugged laser effects that rendered on PC but disappeared in the headset, traced the cause to shaders that do not support single-pass stereo rendering, and replaced them with a compatible fallback

Technical Skills

Languages: Python, C++, C, C#, Java, TypeScript, SQL, R
Computer Vision & Image Processing: OpenCV, PyTorch, torchvision, MediaPipe, NumPy, ModernGL/GLSL
Tools: Git, Docker, Android, Unity XR, React, Next.js, XGBoost, GeoPandas, ArcGIS Pro